Web-based Attacks on Host-Proof Encrypted Storage

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Cloud-based storage services, such as Wuala, and password managers, such as LastPass, are examples of socalled host-proof web applications that aim to protect users from attacks on the servers that host their data. To this end, user data is encrypted on the client and the server is used only as a backup data store. Authorized users may access their data through client-side software, but for ease of use, many commercial applications also offer browser-based interfaces that enable features such as remote access, form-filling, and secure sharing.

We describe a series of web-based attacks on popular host-proof applications that completely circumvent their cryptographic protections. Our attacks exploit standard web application vulnerabilities to expose flaws in the encryption mechanisms, authorization policies, and key management implemented by these applications. Our analysis suggests that host-proofing by itself is not enough to protect users from web attackers, who will simply shift their focus to flaws in client-side interfaces.

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